Quiz 2

Qes1)

```
CREATE TABLE PRODUCT_DIM (
    PRODUCT_ID        NUMBER(6,0)        Primary Key,
    PRODUCT_NAME      VARCHAR2(20 BYTE),
    MANUFACTURER_NAME VARCHAR2(25 BYTE),
    PRODUCT_COLOR     VARCHAR2(25 BYTE) NOT NULL
);
```

Qes2)

```
INSERT INTO PRODUCT_DIM
    (PRODUCT_ID, PRODUCT_NAME, MANUFACTURER_NAME, PRODUCT_COLOR)
VALUES
    (12, 'HTC One', 'HTC', 'Black');

INSERT INTO PRODUCT_DIM
    (PRODUCT_ID, PRODUCT_NAME, MANUFACTURER_NAME, PRODUCT_COLOR)
VALUES
    (13, 'HTC One', 'HTC', 'Gray');

INSERT INTO PRODUCT_DIM
    (PRODUCT_ID, PRODUCT_NAME, MANUFACTURER_NAME, PRODUCT_COLOR)
VALUES
    (14, 'HTC One', 'HTC', 'Blue');

INSERT INTO PRODUCT_DIM
    (PRODUCT_ID, PRODUCT_NAME, MANUFACTURER_NAME, PRODUCT_COLOR)
VALUES
    (15, 'Galaxy A3', 'Samsung', 'White');

INSERT INTO PRODUCT_DIM
    (PRODUCT_ID, PRODUCT_NAME, MANUFACTURER_NAME, PRODUCT_COLOR)
VALUES
    (16, 'Galaxy A3', 'Samsung', 'Black');

INSERT INTO PRODUCT_DIM
    (PRODUCT_ID, PRODUCT_NAME, MANUFACTURER_NAME, PRODUCT_COLOR)
VALUES
    (17, 'Galaxy A3', 'Samsung', 'Blue');

INSERT INTO PRODUCT_DIM
    (PRODUCT_ID, PRODUCT_NAME, MANUFACTURER_NAME, PRODUCT_COLOR)
VALUES
    (18, 'Galaxy A5', 'Samsung', 'White');

INSERT INTO PRODUCT_DIM
    (PRODUCT_ID, PRODUCT_NAME, MANUFACTURER_NAME, PRODUCT_COLOR)
VALUES
    (19, 'Galaxy A5', 'Samsung', 'Black');

INSERT INTO PRODUCT_DIM
    (PRODUCT_ID, PRODUCT_NAME, MANUFACTURER_NAME, PRODUCT_COLOR)
```

```
VALUES
    (20, 'Galaxy A5', 'Samsung', 'Blue');

INSERT INTO PRODUCT_DIM
    (PRODUCT_ID, PRODUCT_NAME, MANUFACTURER_NAME, PRODUCT_COLOR)
VALUES
    (21, 'IPHONE 6', 'Apple', 'Gold');

INSERT INTO PRODUCT_DIM
    (PRODUCT_ID, PRODUCT_NAME, MANUFACTURER_NAME, PRODUCT_COLOR)
VALUES
    (22, 'IPHONE 6', 'Apple', 'white');

COMMIT;
```

Qes3)

```
CREATE TABLE EMPLOYEE_DIM (
    EMPLOYEE_ID   NUMBER (6,0)      Primary Key,
    FIRST_NAME    VARCHAR2(20)    NULL,
    LAST_NAME     VARCHAR2(25)    NOT NULL,
    EMAIL         VARCHAR2(25)    NOT NULL,
    PHONE_NUMBER VARCHAR2(20)    NULL,
    HIRE_DATE     DATE            NOT NULL
);
```

Qes4)

```
INSERT INTO EMPLOYEE_DIM
SELECT
    EMPLOYEE_ID,
    FIRST_NAME,
    LAST_NAME,
    EMAIL,
    PHONE_NUMBER,
    HIRE_DATE
FROM HR. EMPLOYEES;
```

Qes5)

```
CREATE TABLE CUSTOMER_DIM (
    CUSTOMER_ID      NUMBER (6,0)      Primary Key,
    FIRST_NAME       VARCHAR2(20)    NULL,
    LAST_NAME        VARCHAR2(25)    NOT NULL,
    EMAIL            VARCHAR2(25)    NOT NULL,
    PHONE_NUMBER     VARCHAR2(20)    NULL,
    ACTIVATION_DATE DATE            NOT NULL
);
```

Qes6)

```
INSERT INTO CUSTOMER_DIM
    (CUSTOMER_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, ACTIVATION_DATE)
VALUES
    (1001, 'AHMED', 'MOHAMED', 'AHMED@GMAIL.COM', '3243546',
     TO_DATE('13-03-10', 'DD-MM-RR'));

INSERT INTO CUSTOMER_DIM
    (CUSTOMER_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, ACTIVATION_DATE)
VALUES
    (1002, 'MOHAMED', 'AHMED', 'MAHMED@GMAIL.COM', '3455436',
     TO_DATE('15-07-15', 'DD-MM-RR'));

INSERT INTO CUSTOMER_DIM
    (CUSTOMER_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, ACTIVATION_DATE)
VALUES
    (1003, 'ALI', 'MAHMOUD', 'ALIMAHMOUD@GMAIL.COM', '5645675',
     TO_DATE('12-10-05', 'DD-MM-RR'));

INSERT INTO CUSTOMER_DIM
    (CUSTOMER_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, ACTIVATION_DATE)
VALUES
    (1004, 'MAHMOUD', 'MOHAMED', 'MMOHAMED@GMAIL.COM', '4565467',
     TO_DATE('03-02-12', 'DD-MM-RR'));

INSERT INTO CUSTOMER_DIM
    (CUSTOMER_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, ACTIVATION_DATE)
VALUES
    (1005, 'KAREEM', 'ZAKI', 'KAREEMZ@GMAIL.COM', '5465477',
     TO_DATE('12-09-13', 'DD-MM-RR'));

INSERT INTO CUSTOMER_DIM
    (CUSTOMER_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, ACTIVATION_DATE)
VALUES
    (1006, 'AMR', 'SHABAN', 'ASH@GMAIL.COM', '4356657',
     TO_DATE('12-12-12', 'DD-MM-RR'));

INSERT INTO CUSTOMER_DIM
    (CUSTOMER_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, ACTIVATION_DATE)
VALUES
    (1007, 'OMAR', 'ZYAD', 'OMZYAD@GMAIL.COM', '12456044',
     TO_DATE('16-09-15', 'DD-MM-RR'));

INSERT INTO CUSTOMER_DIM
    (CUSTOMER_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, ACTIVATION_DATE)
VALUES
    (1008, 'MOHAMED', 'KAREEM', 'MKAREEM@GMAIL.COM', '13459593',
     TO_DATE('17-08-01', 'DD-MM-RR'));

INSERT INTO CUSTOMER_DIM
    (CUSTOMER_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, ACTIVATION_DATE)
VALUES
    (1009, 'IBRAHIM', 'SALEH', 'IBRAHIMSALEH@GAIL.COM', '57979483',
     TO_DATE('30-10-14', 'DD-MM-RR'));

INSERT INTO CUSTOMER_DIM
```

```
    (CUSTOMER_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, ACTIVATION_DATE)
VALUES
    (1010, 'MOHAMED', 'IBRAHIM', 'MOHAMEDIBRAH@GMAIL.COM', '39469684',
     TO_DATE('14-02-02', 'DD-MM-RR'));

COMMIT;
```

Qes7)

```
CREATE TABLE CAIRO_SHOP_SALES (
    REQ_ID      NUMBER (6,0) NOT NULL,
    EMP_ID      NUMBER (6,0),
    CUSTOMER_ID NUMBER (6,0),
    PRODUCT_ID NUMBER (6,0),
    PRICE       NUMBER (15,0),
    ENTRY_DATE TIMESTAMP (6) DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT CAIRO_PRICE_CHK
        CHECK (PRICE <= 15000),

    CONSTRAINT CAIRO_EMP_FK
        FOREIGN KEY (EMP_ID)
        REFERENCES EMPLOYEE_DIM (EMPLOYEE_ID),

    CONSTRAINT CAIRO_CUSTOMER_FK
        FOREIGN KEY (CUSTOMER_ID)
        REFERENCES CUSTOMER_DIM (CUSTOMER_ID),

    CONSTRAINT CAIRO_PRODUCT_FK
        FOREIGN KEY (PRODUCT_ID)
        REFERENCES PRODUCT_DIM (PRODUCT_ID)
);
```

```
CREATE TABLE GIZA_SHOP_SALES (
    REQ_ID      NUMBER (6,0) NOT NULL,
    EMP_ID      NUMBER (6,0),
    CUSTOMER_ID NUMBER (6,0),
    PRODUCT_ID NUMBER(6,0),
    PRICE       NUMBER (15,0),
    ENTRY_DATE TIMESTAMP (6) DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT GIZA_PRICE_CHK
        CHECK (PRICE <= 15000),

    CONSTRAINT GIZA_EMP_FK
        FOREIGN KEY (EMP_ID)
        REFERENCES EMPLOYEE_DIM (EMPLOYEE_ID),

    CONSTRAINT GIZA_CUSTOMER_FK
        FOREIGN KEY (CUSTOMER_ID)
        REFERENCES CUSTOMER_DIM (CUSTOMER_ID),

    CONSTRAINT GIZA_PRODUCT_FK
```

```
        FOREIGN KEY (PRODUCT_ID)
        REFERENCES PRODUCT_DIM (PRODUCT_ID)
);
```

Qes8)

```
INSERT INTO CAIRO_SHOP_SALES
    (REQ_ID, EMP_ID, CUSTOMER_ID, PRODUCT_ID, PRICE)
VALUES
    (1, 101, 1001, 12, 2500),
    (2, 101, 1002, 15, 3000),
    (3, 102, 1002, 13, 5999),
    (4, 103, 1005, 17, 11000),
    (5, 102, 1003, 14, 3999),
    (6, 104, 1004, 20, 7999),
    (7, 105, 1007, 18, 4999),
    (8, 104, 1009, 16, 11000),
    (9, 105, 1001, 19, 4500),
    (10, 103, 1008, 21, 3560);
```

```
INSERT INTO GIZA_SHOP_SALES
    (REQ_ID, EMP_ID, CUSTOMER_ID, PRODUCT_ID, PRICE)
VALUES
    (11, 106, 1006, 18, 4999),
    (12, 106, 1008, 19, 4500),
    (13, 108, 1002, 21, 3560),
    (14, 107, 1010, 17, 11000),
    (15, 110, 1003, 16, 11000),
    (16, 109, 1004, 12, 2500),
    (17, 109, 1005, 15, 3000),
    (18, 108, 1007, 13, 5999),
    (19, 107, 1009, 14, 3999),
    (20, 110, 1001, 20, 7999);
```

Qes9)

```
CREATE VIEW SALES_FACT AS
SELECT c.*,
'CAIRO' AS BRANCH
FROM CAIRO_SHOP_SALES c
UNION ALL
SELECT g.*,
'GIZA' AS BRANCH
 FROM GIZA_SHOP_SALES g;
```

Qes10)

- first

```
CREATE VIEW EMPLOYEE_OF_THE_MONTH AS
SELECT
    E. EMPLOYEE_ID AS "Employee ID",
```

```
    (E. FIRST_NAME||' '||E.LAST_NAME) AS "Full Name",
    COUNT (S. PRODUCT_ID) AS "Number of Products Sold"
FROM SALES_FACT S
JOIN EMPLOYEE_DIM E
    ON S.EMP_ID = E. EMPLOYEE_ID
GROUP BY
    E. EMPLOYEE_ID,
    "Full Name"
ORDER BY
    COUNT(S. PRODUCT_ID) DESC;
```

- Second

```
SELECT *
FROM (
    SELECT
        C.CUSTOMER_ID AS "Customer ID",
        (C. FIRST_NAME ||' '|| C. LAST_NAME) AS "Full Name",
        C.EMAIL AS "Email",
        SUM (S. PRICE) AS "Total number of purchases"
    FROM SALES_FACT S
    JOIN CUSTOMER_DIM C
        ON S. CUSTOMER_ID = C. CUSTOMER_ID
    GROUP BY
        C.CUSTOMER_ID,
        "Full Name",
        C.EMAIL
    ORDER BY SUM (S. PRICE) DESC
)
WHERE ROWNUM = 1;
```

- Third

```
SELECT DISTINCT
    C.CUSTOMER_ID AS "Customer ID",
    (C. FIRST_NAME ||' '||C.LAST_NAME) AS "Full Name",
    C.EMAIL AS "Email",
    C.ACTIVATION_DATE AS "Activation Date",
    P. PRODUCT_NAME AS "Mobile Name"
FROM SALES_FACT S
JOIN CUSTOMER_DIM C
    ON S. CUSTOMER_ID = C. CUSTOMER_ID
JOIN PRODUCT_DIM P
    ON S. PRODUCT_ID = P. PRODUCT_ID
WHERE P. MANUFACTURER_NAME LIKE '%Samsung%';
```

- Fourth

```
SELECT
    P. PRODUCT_ID,
    P. PRODUCT_NAME,
    P. MANUFACTURER_NAME,
    P. PRODUCT_COLOR
FROM PRODUCT_DIM P
WHERE P. PRODUCT_ID IN (
    SELECT PRODUCT_ID
```

```
    FROM CAIRO_SHOP_SALES
)
AND P. PRODUCT_ID IN (
    SELECT PRODUCT_ID
    FROM GIZA_SHOP_SALES
);
```

- Fifth

```
SELECT DISTINCT
    C.CUSTOMER_ID AS "Customer ID",
    (C. FIRST_NAME||' '||C.LAST_NAME) AS "Full Name",
    C.EMAIL AS "Email",
    P. PRODUCT_NAME AS "Mobile Name",
    P. MANUFACTURER_NAME AS "Mobile Type"
FROM CUSTOMER_DIM C
JOIN SALES_FACT S
    ON C. CUSTOMER_ID = S. CUSTOMER_ID
JOIN PRODUCT_DIM P
    ON S. PRODUCT_ID = P. PRODUCT_ID
WHERE C. CUSTOMER_ID IN (
    SELECT CUSTOMER_ID
    FROM CAIRO_SHOP_SALES
)
AND C. CUSTOMER_ID IN (
    SELECT CUSTOMER_ID
    FROM GIZA_SHOP_SALES
)
```

Qes11)

```
CREATE USER BI_DEVELOPER
IDENTIFIED BY Str0ng_Pwd#2026;
```

```
GRANT CREATE SESSION TO BI_DEVELOPER;

GRANT SELECT ON SALES_FACT TO BI_DEVELOPER;

GRANT SELECT ON CUSTOMER_DIM TO BI_DEVELOPER;

GRANT SELECT ON PRODUCT_DIM TO BI_DEVELOPER;
```